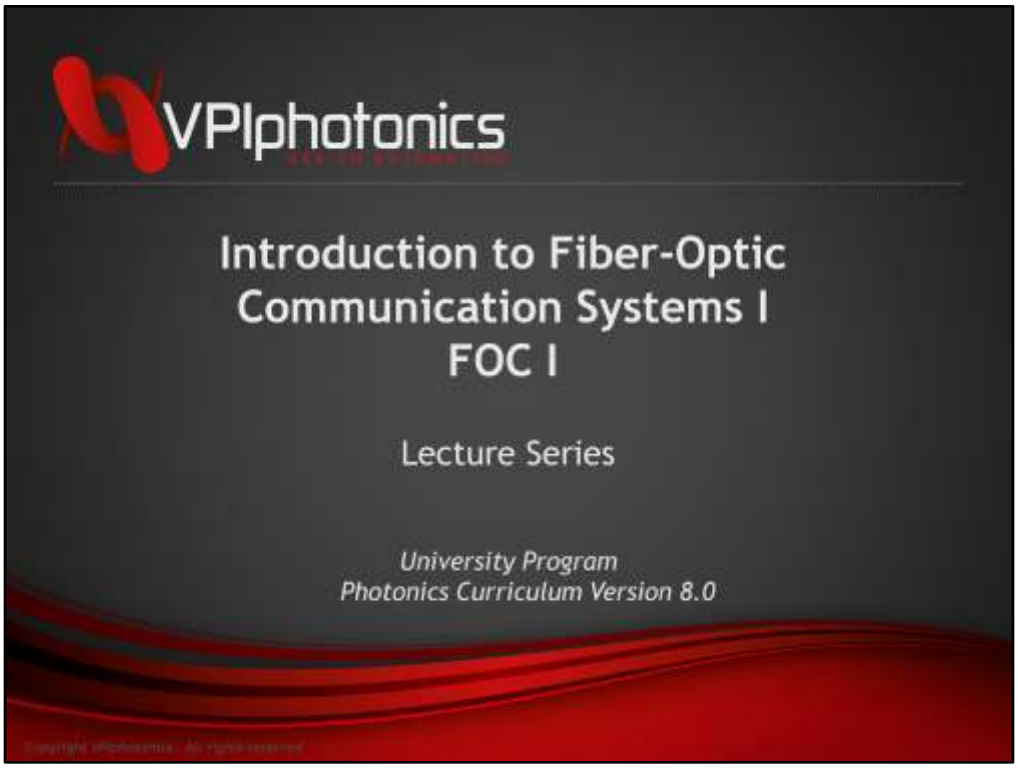
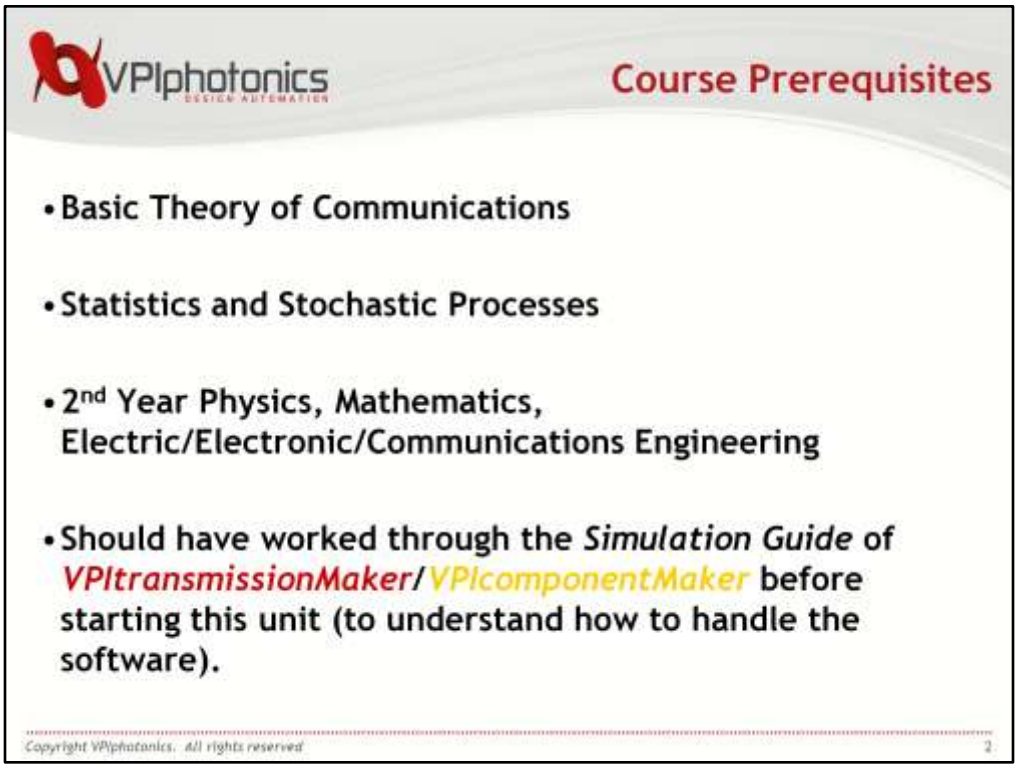
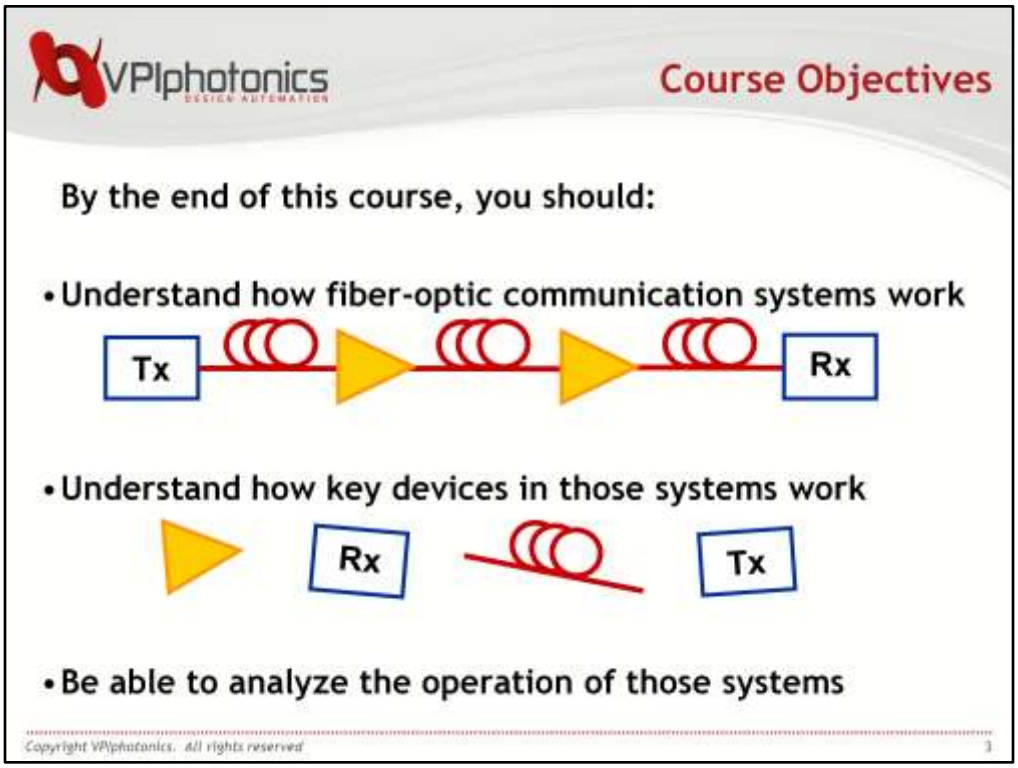
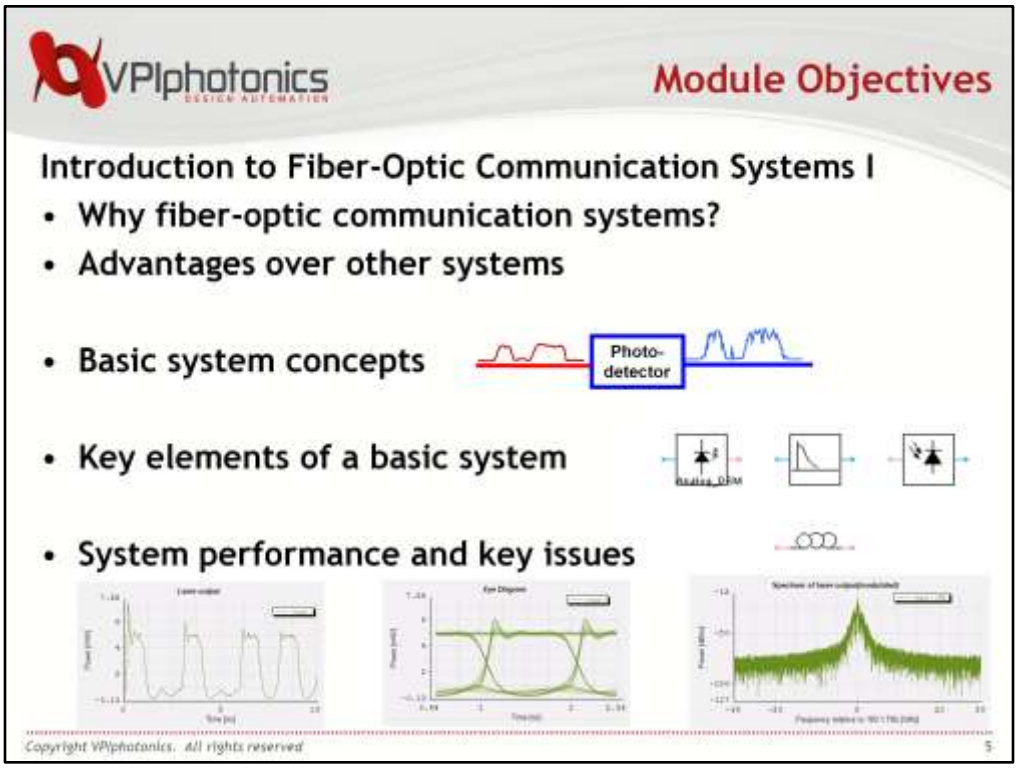

VPIphotonics
Introduction to Fiber-Optic Communication Systems I
FOC I
Lecture Series
University Program
Photonics Curriculum Version 8.0



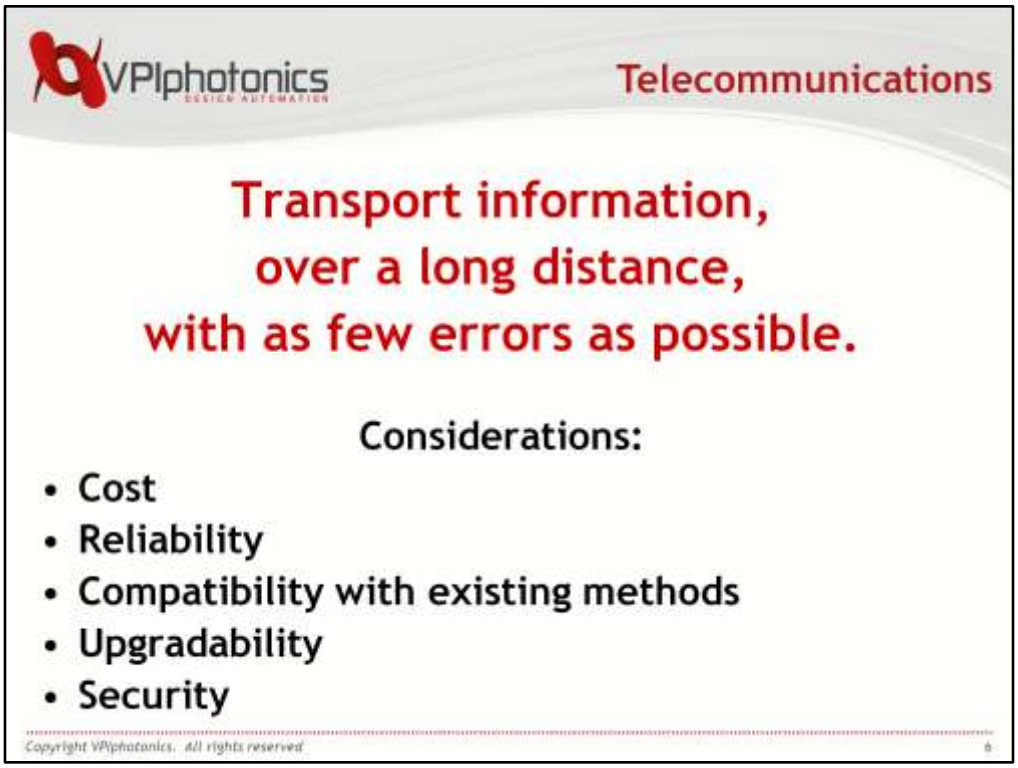
Statement of the prerequisites that students should have before doing this course. Please modify this slide to suit the specific requirements of your course.



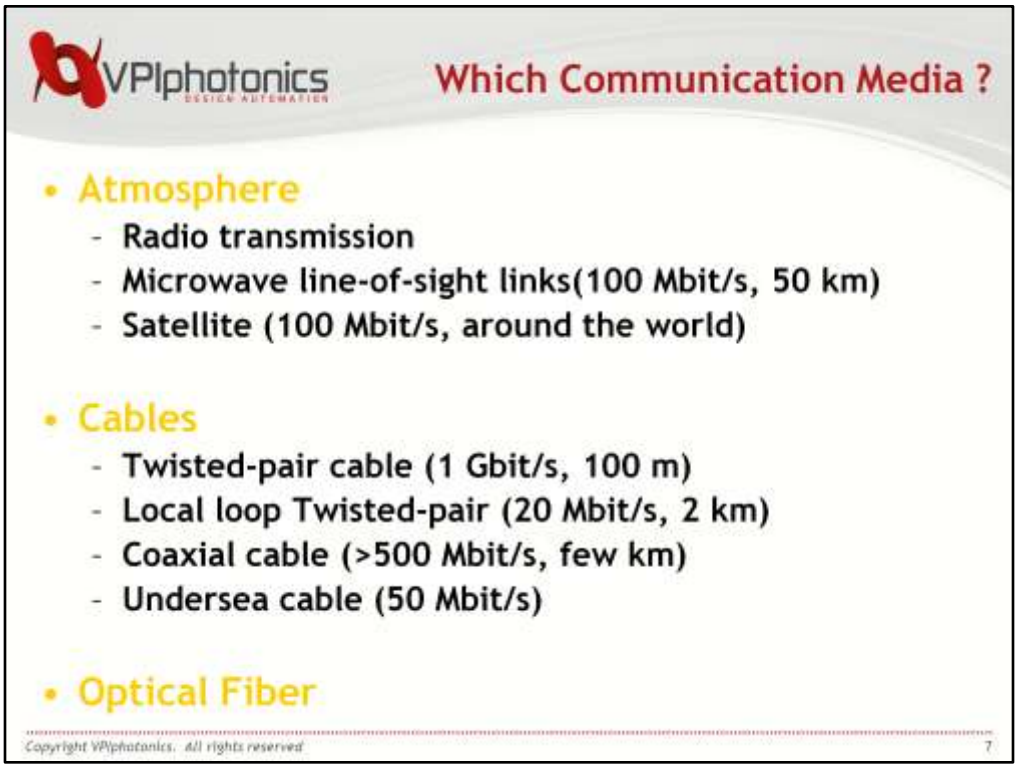
Interactively explore the workings of a basic fiber-optic communications system via a computer model



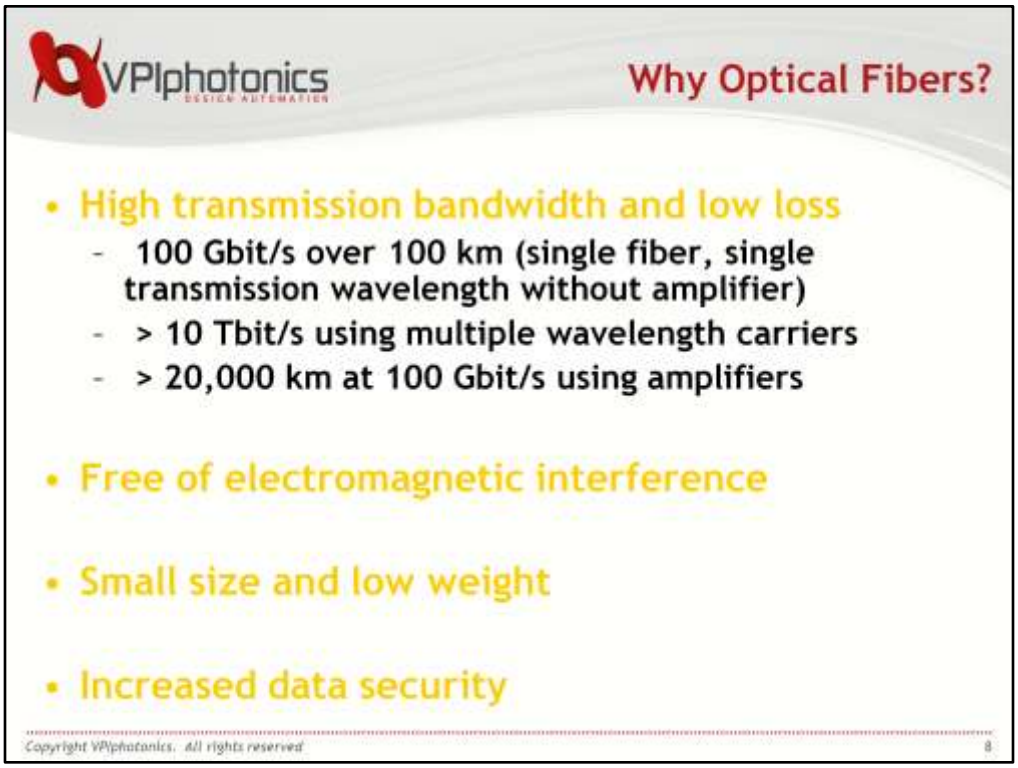
An outline of the objectives of this module.



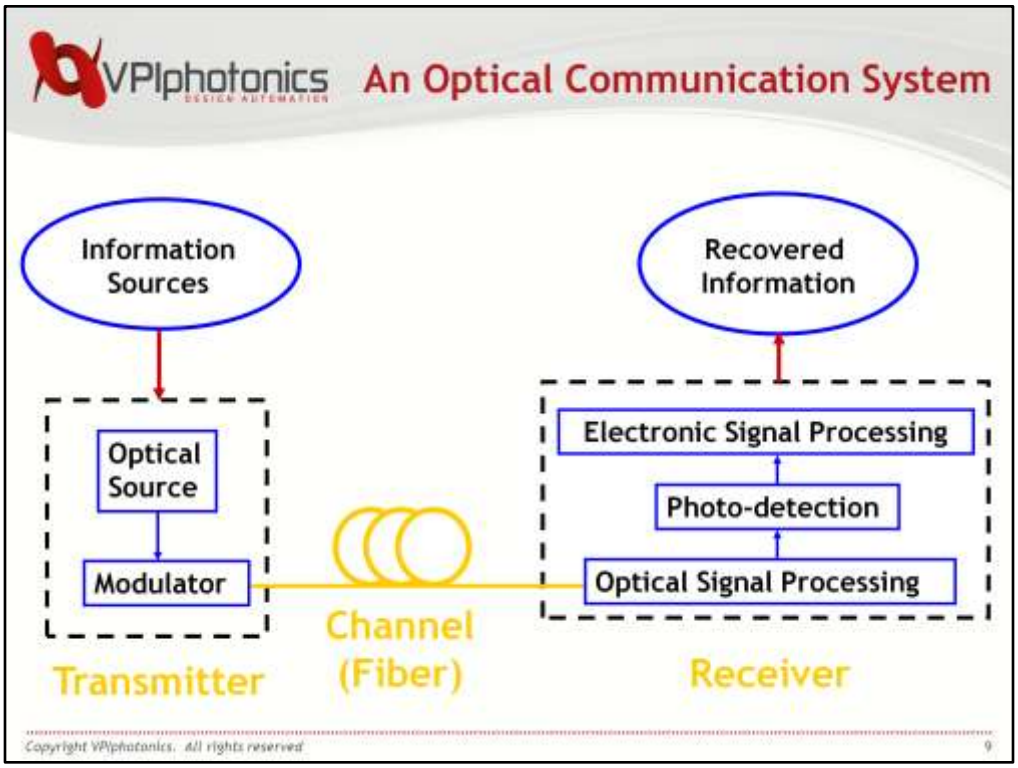
Key considerations for a telecommunication system.



A list of existing communication media, along with some numbers to show the capacity and distance typically achieved (may need to be updated as new advances are made in the respective communication medium).



Focusing on the why optical fibers are used, and providing some numbers and reasons for justification.



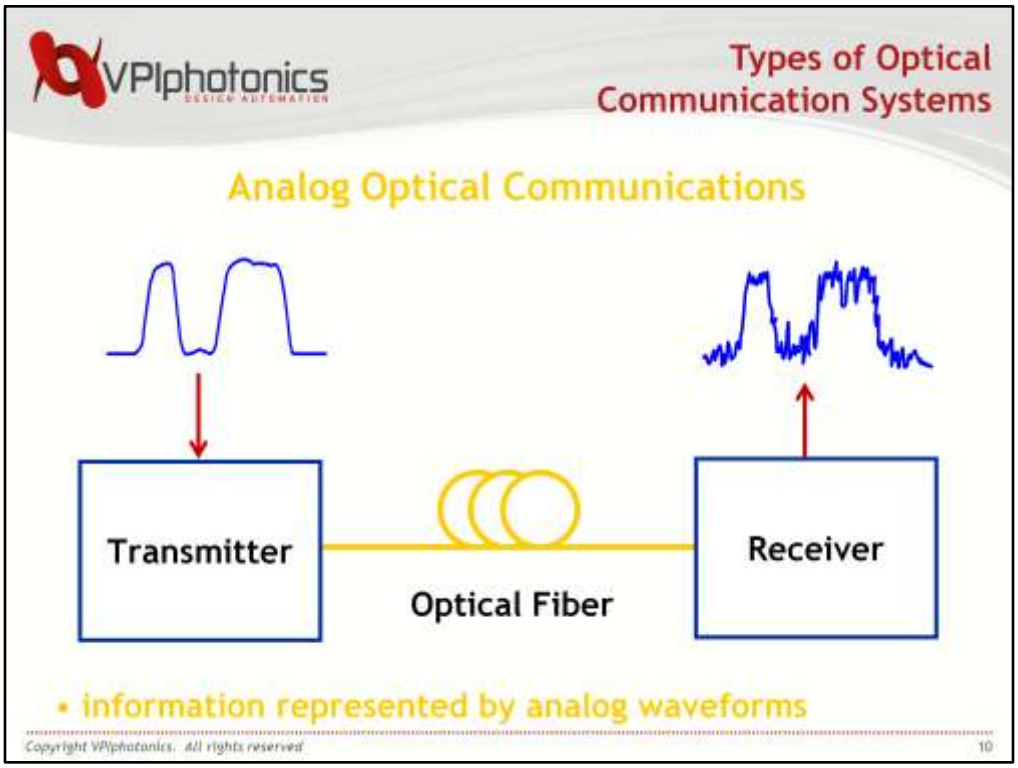
An Optical Communication System

In principle, can be represented by a generic block diagram of a communication system – Transmitter, Channel and Receiver

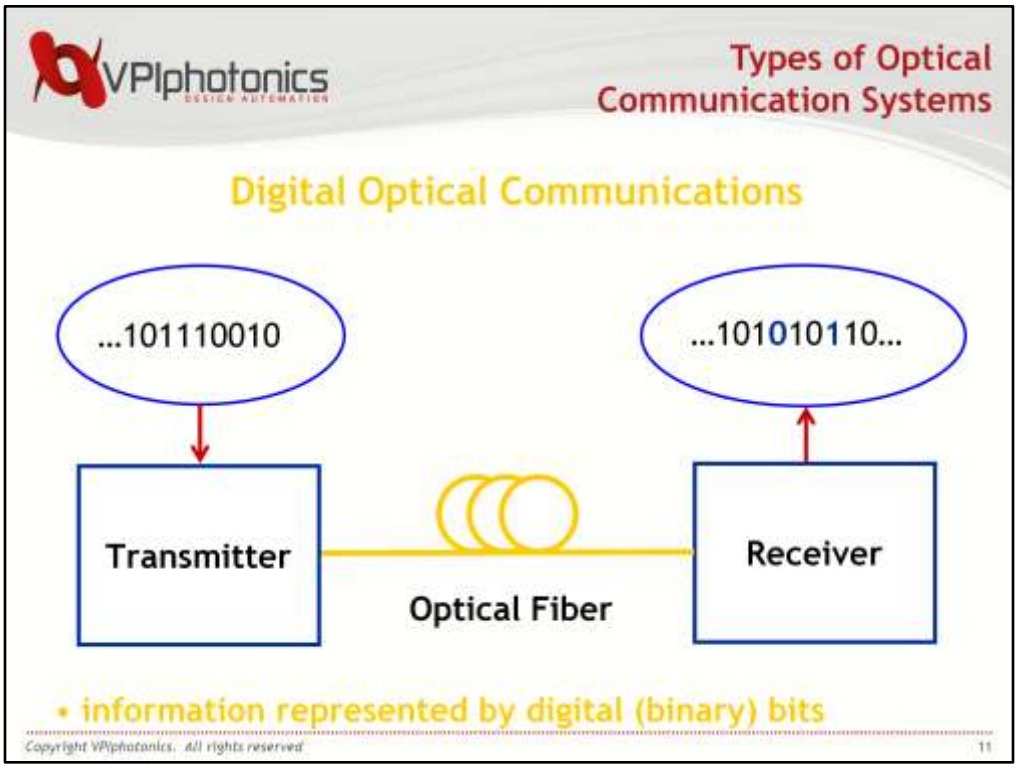
1) Source of information feeds into the transmitter
2) Information is encoded, modulated and transmitted over the channel
3) Receiver demodulates, decodes and recover the transmitted information

The devil (difference) is in the details –

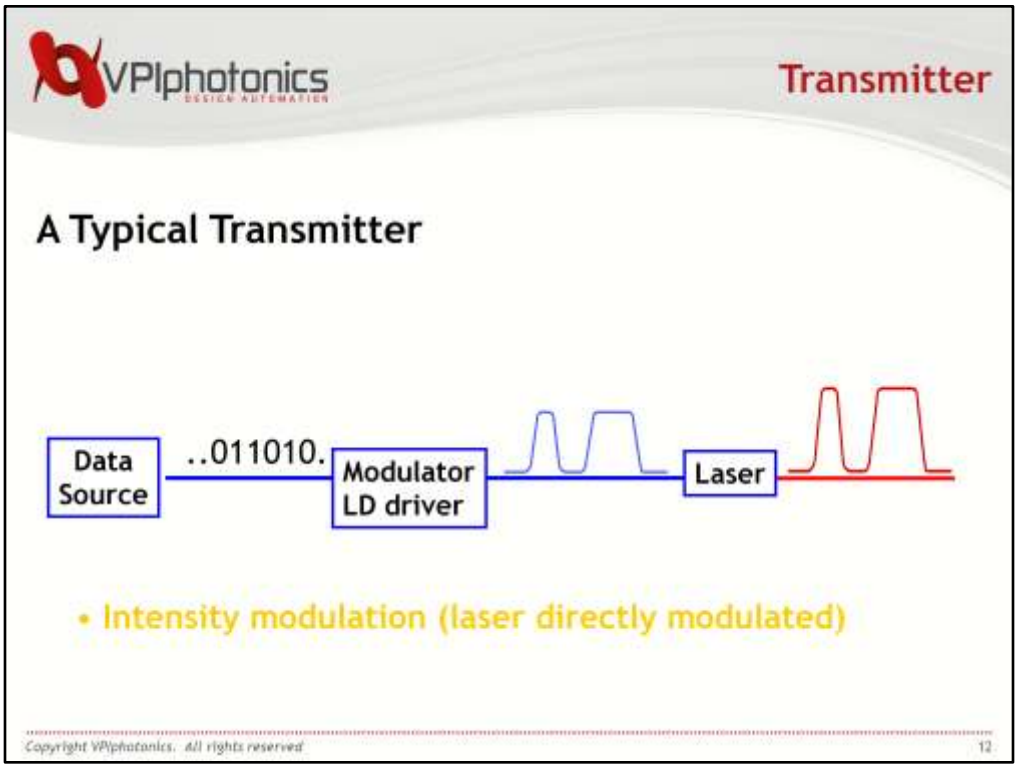
1) Inside the transmitter is an optical source, and some means to optically modulate the information
2) In the receiver, there may be some optical signal processing prior to photodetection of the received optical signal. There will be further post-processing in the electrical domain to recover the transmitted information.



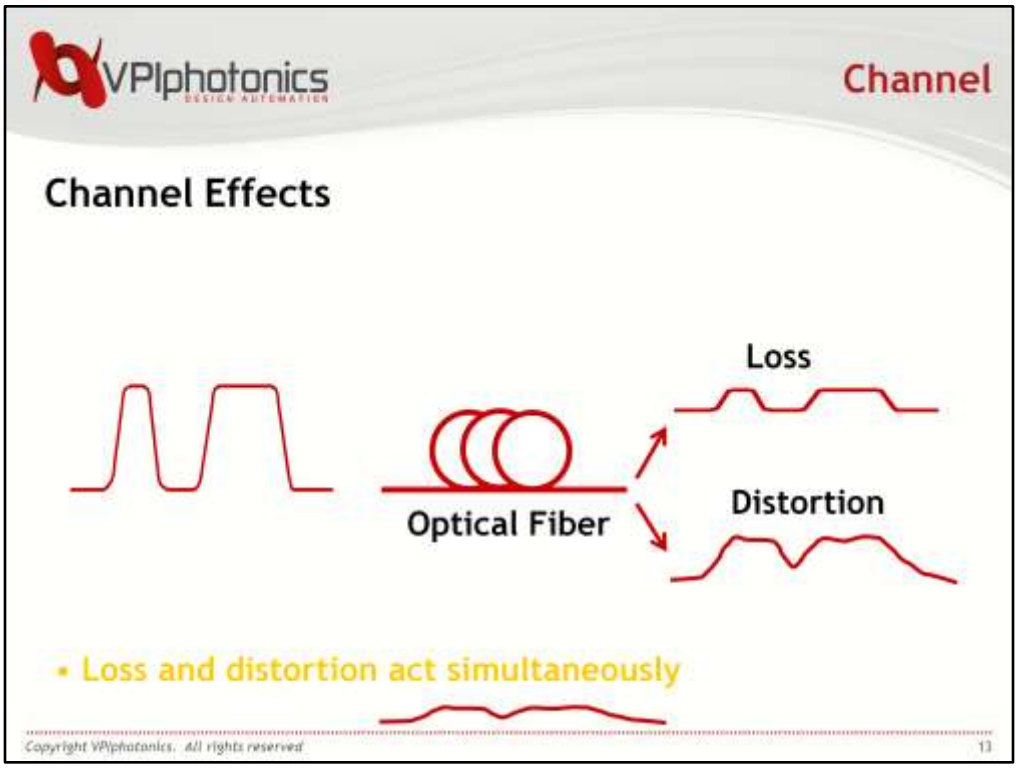
The received signal is distorted and corrupted by noise.



In a digital system, the effect of noise and distortion results in bit errors (as highlighted in red/bold in the digital bit stream at the receiver).



Data source: Usually a digital binary signal, but may also be an analog signal in some applications.
Coder: Transform signal to a desired format for modulation of an optical carrier wave. Examples are Non-Return-to-Zero (NRZ), Return-to-Zero (RZ), Quadrature Amplitude Modulation (QAM). Additional coders may be used to perform forward error correction coding.
Laser: The source of the optical carrier wave, which is modulated by the signal from the coder. Basically performs conversion of encoded signal from the electrical domain to the optical domain. LEDs may also be used for this purpose.



Fiber loss varies with the wavelength of the signal, but is much lower than that of the cables used to transmit electrical signals. Therefore, the maximum distance of transmission before the optical signals require amplification is large (up to several hundred kilometers). Thus, fibre-optic transmission is favoured, and predominates, in transoceanic and long distance land-based point-to-point communication systems.

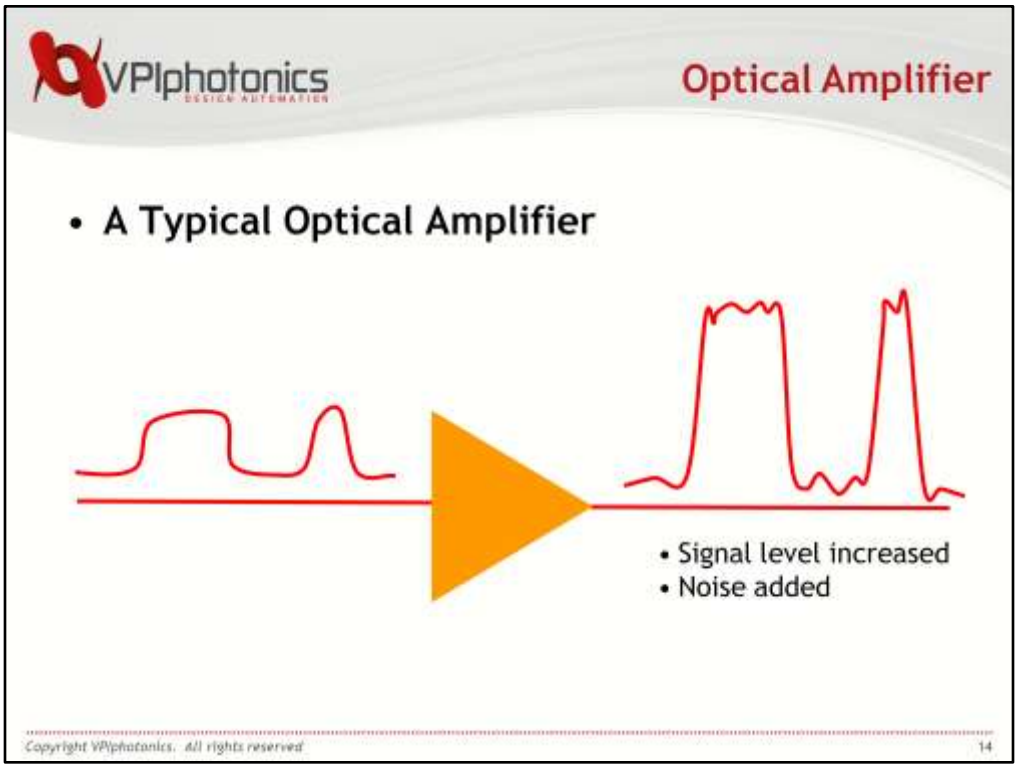
Basically, simultaneous transmission of many optical signals of different wavelengths, coupled with the very high data rates of each individual signal means that optical fibers are capable of an enormous throughput of data (of up to multiple Terabits per second).

Signals of wide wavelength (λ) range can be transmitted over fiber
Eg. 850 nm, 1300 nm, 1530nm – 1580nm

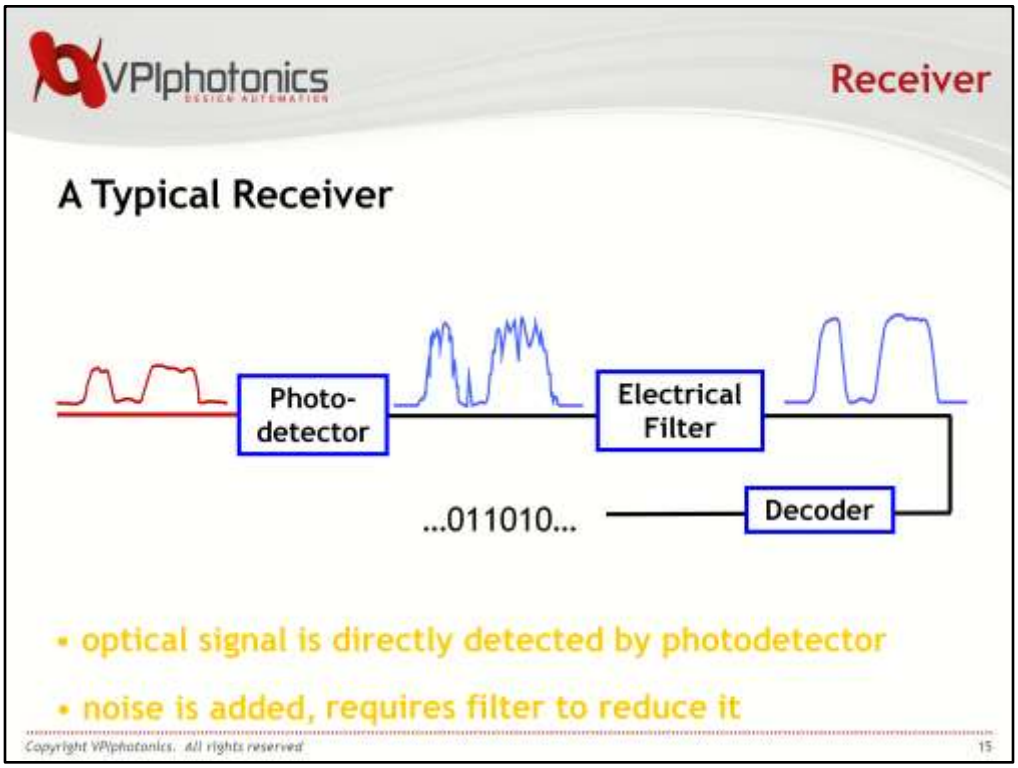
Simultaneous transmission of many wavelengths
Up to several hundred channels

Low loss signal transmission
(~0.2 dB/km at 1550 nm)

Very high speed optical signals
Up to 40 Gigabits per second (Gb/s)



Noise is added by an optical amplifier, in addition to optical signal amplification.



A Typical Receiver Consists of:

Photodetector - optical to electrical conversion of signal
Electrical Filter - reduce noise
Equipment for recovery of transmitted data- decoder, error correction, clock recovery etc.
Test and measurement equipment for signal quality monitoring.

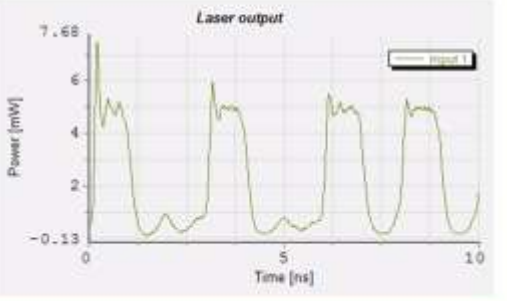
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System Performance

- **Signal waveform in time domain**

Equipment	Resolution
Oscilloscope:	~ 100 ps
Sampling Scope	~ 20 ps
Streak Camera	~ 2 ps
Auto-correlator	~ 10 - 100 fs

Typical Result of SignalAnalyzer (Scope view) →



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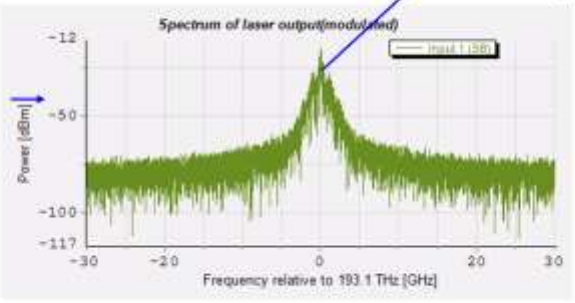
The detected signal waveform can be examined to assess the performance of the system. Various test and measurement equipment are used to examine

Evaluating System Performance

- **Optical spectrum**

Lasing wavelength
Linewidth
Noise
:

Typical Result of SignalAnalyzer (OSA view)



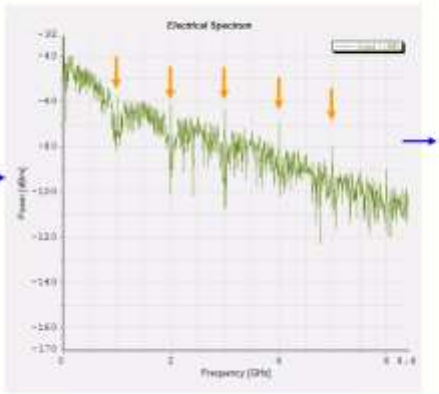
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Evaluating System Performance

- **RF Spectrum**

RF components at f_o, $2f_o$, $3f_o$...

Typical Result of SignalAnalyzer (RFSA view)



Repetition rate
Phase noise
RIN
⋮

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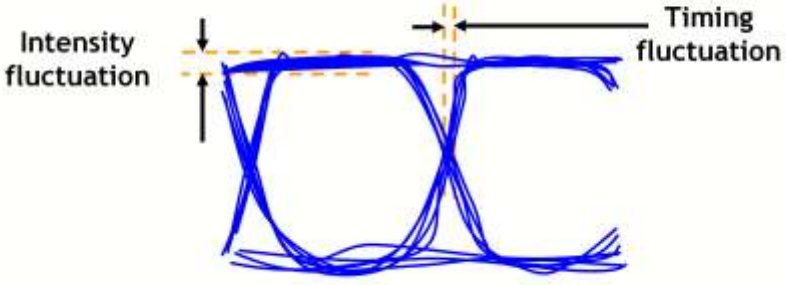
VPIphotonics DESIGN AUTOMATION

Evaluating System Performance

- **Eye diagram**

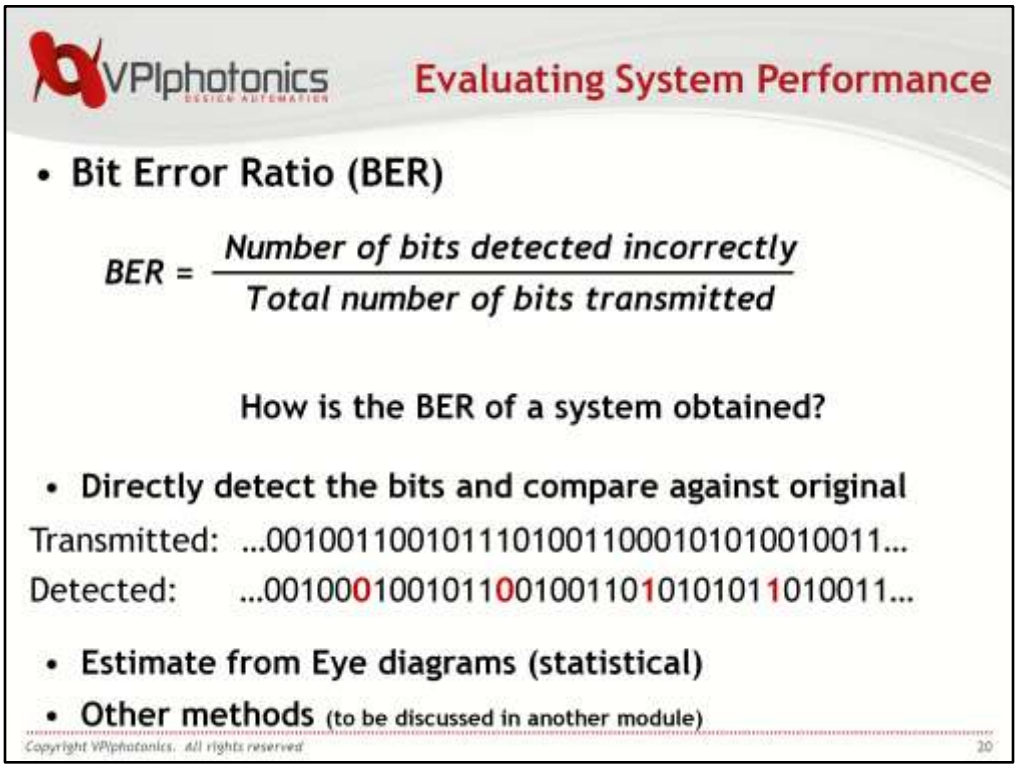


Detected bit stream waveform



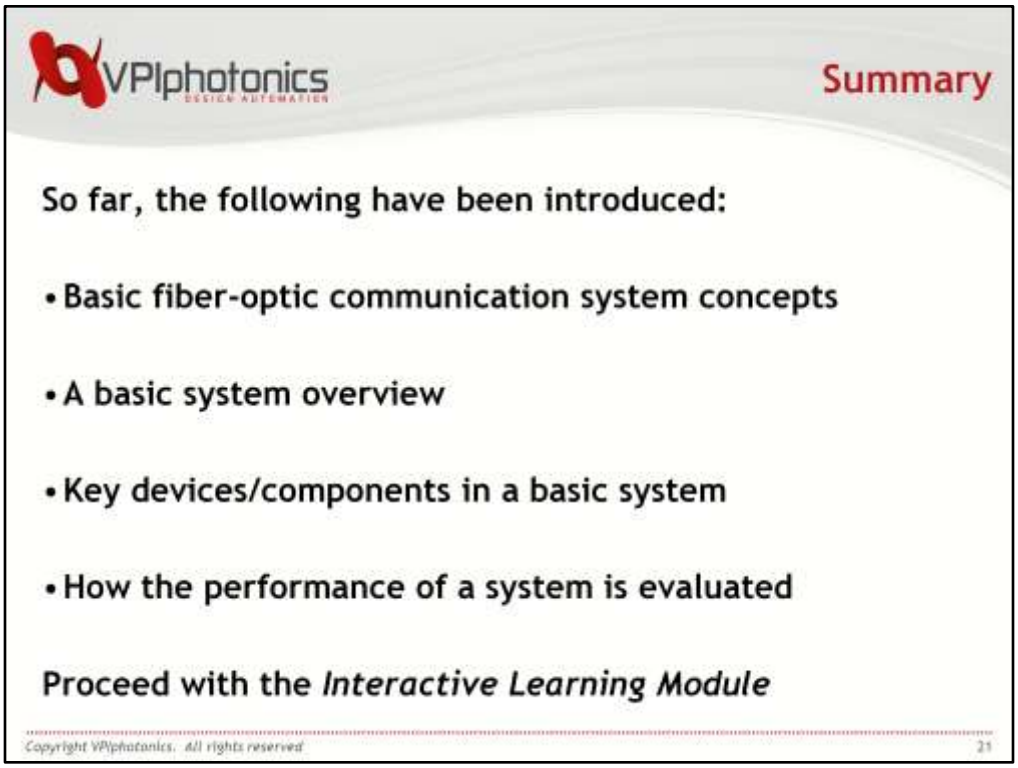
Bits overlaid to form Eye diagram

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Note to instructor:

This slide provides a “high-level” introduction to the bit error ratio (BER) as a system performance measure. The detailed discussion of BER will be addressed in a separate module. The reason for bringing up BER in the introductory lecture is so that it can be used to compare the performance of different transmission links that are affected by different amounts of dispersion, loss, laser chirp etc.



Summary of the lecture part of the module.